Math 443 Final Project

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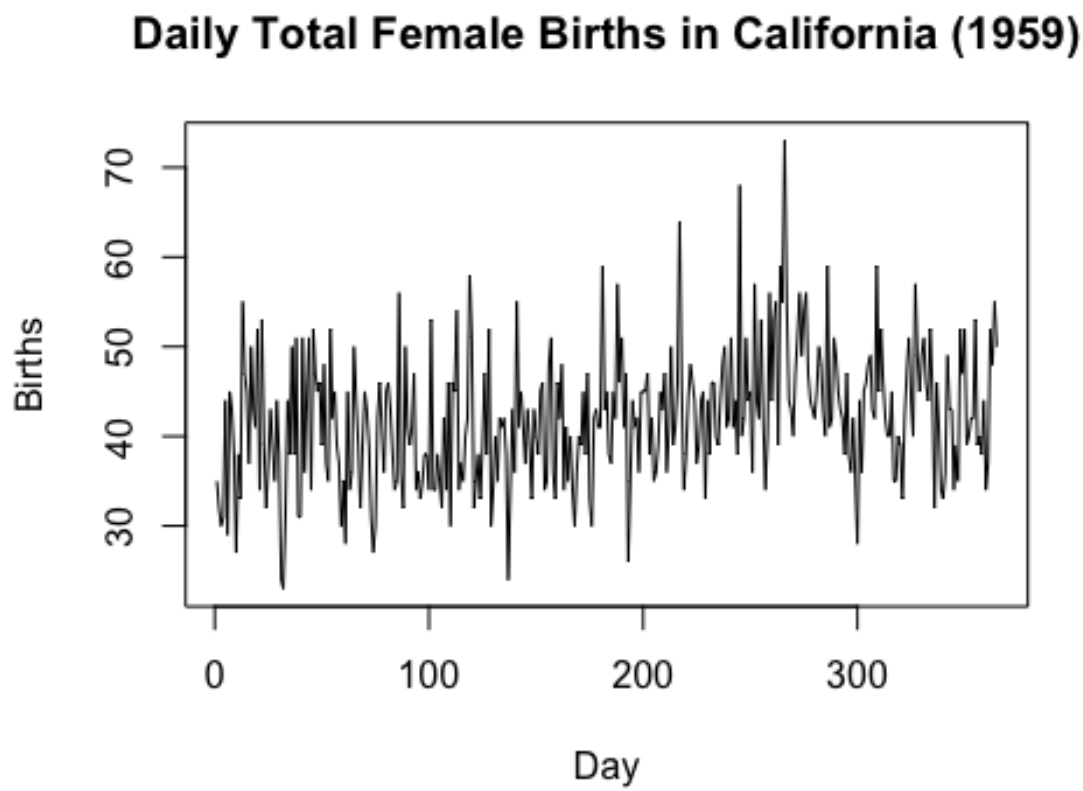
Time Series Project

```
library(readr)

df <- read_csv("/Users/andreiasilvadossantos/Desktop/daily-total-female-
births-in-cal.csv")
```

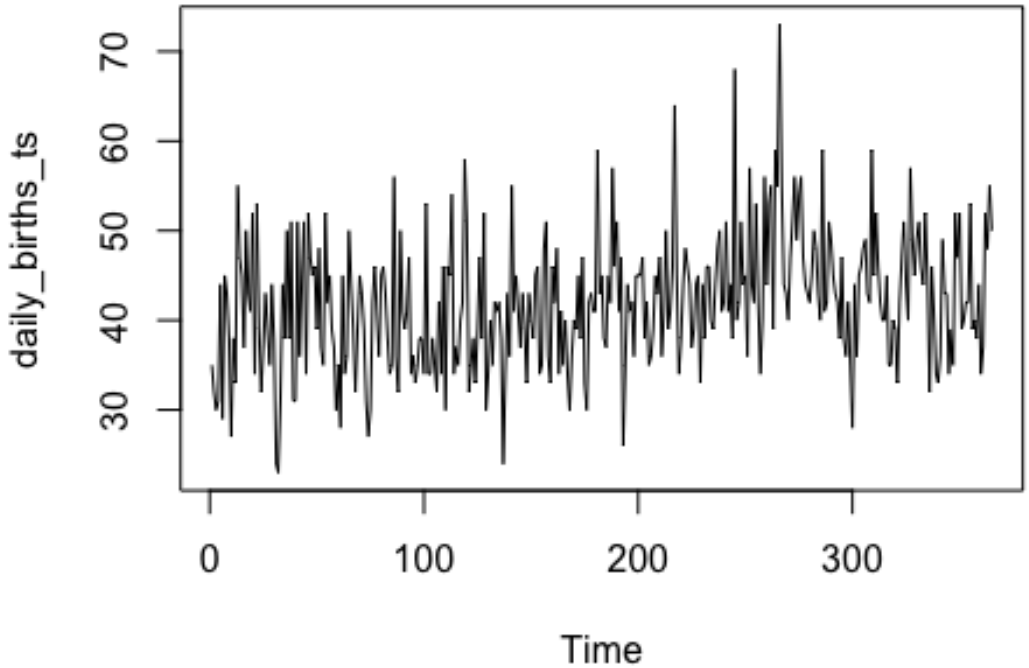
```
## Rows: 365 Columns: 2
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr (1): Date
## dbl (1): Daily total female births in California, 1959
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
daily_births <- df$`Daily total female births in California, 1959`


plot(daily_births, type = "l",
     xlab = "Day",
     ylab = "Births",
     main = "Daily Total Female Births in California (1959)")
```



Summary statistics

```
mean(daily_births)

## [1] 41.98082

sd(daily_births)

## [1] 7.348257

min(daily_births)

## [1] 23

max(daily_births)

## [1] 73

daily_births_ts = ts(daily_births)
plot(daily_births_ts)
```



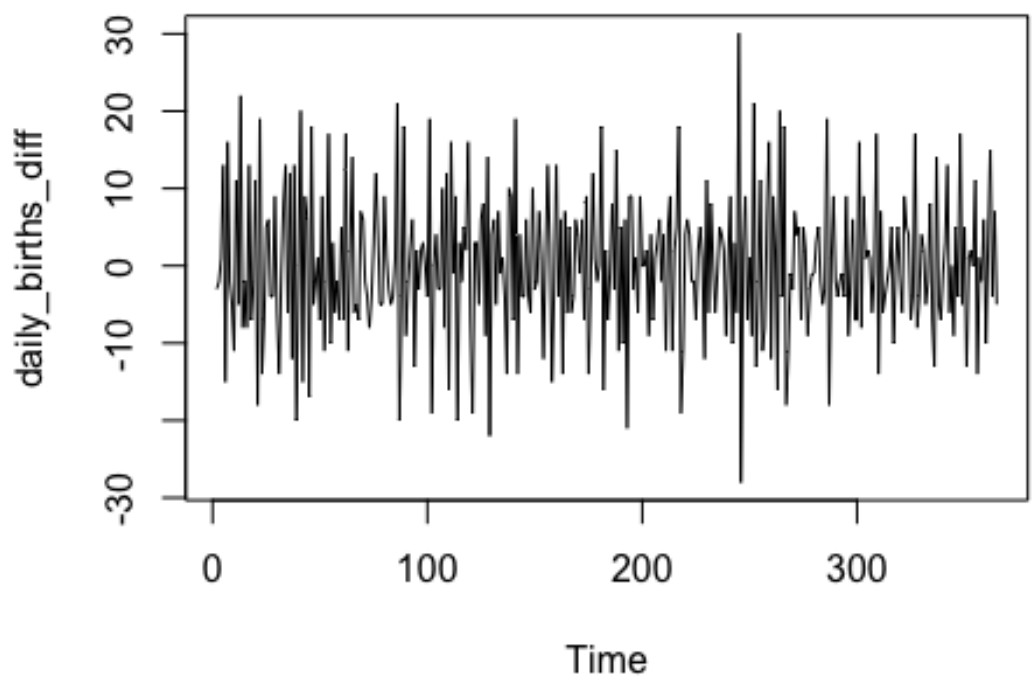
ADF / KPSS tests

```
library(tseries)

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo

adf.test(daily_births_ts)

## Warning in adf.test(daily_births_ts): p-value smaller than printed p-value

##
##  Augmented Dickey-Fuller Test
##
## data:  daily_births_ts
## Dickey-Fuller = -5.1042, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary

kpss.test(daily_births_ts)

## Warning in kpss.test(daily_births_ts): p-value smaller than printed p-
value
```

```
##
##  KPSS Test for Level Stationarity
##
## data:  daily_births_ts
## KPSS Level = 1.8277, Truncation lag parameter = 5, p-value = 0.01
```

Diffeerencing first-order -stabilize the mean -make the series stationary -prepare the data for ARMA / ARIMA modeling

```
daily_births_diff = diff(daily_births_ts)
plot(daily_births_diff)
```



ADF / KPSS of Diffeerencing first-order

```
adf.test(daily_births_diff)

## Warning in adf.test(daily_births_diff): p-value smaller than printed p-
value

##
##  Augmented Dickey-Fuller Test
##
## data:  daily_births_diff
```

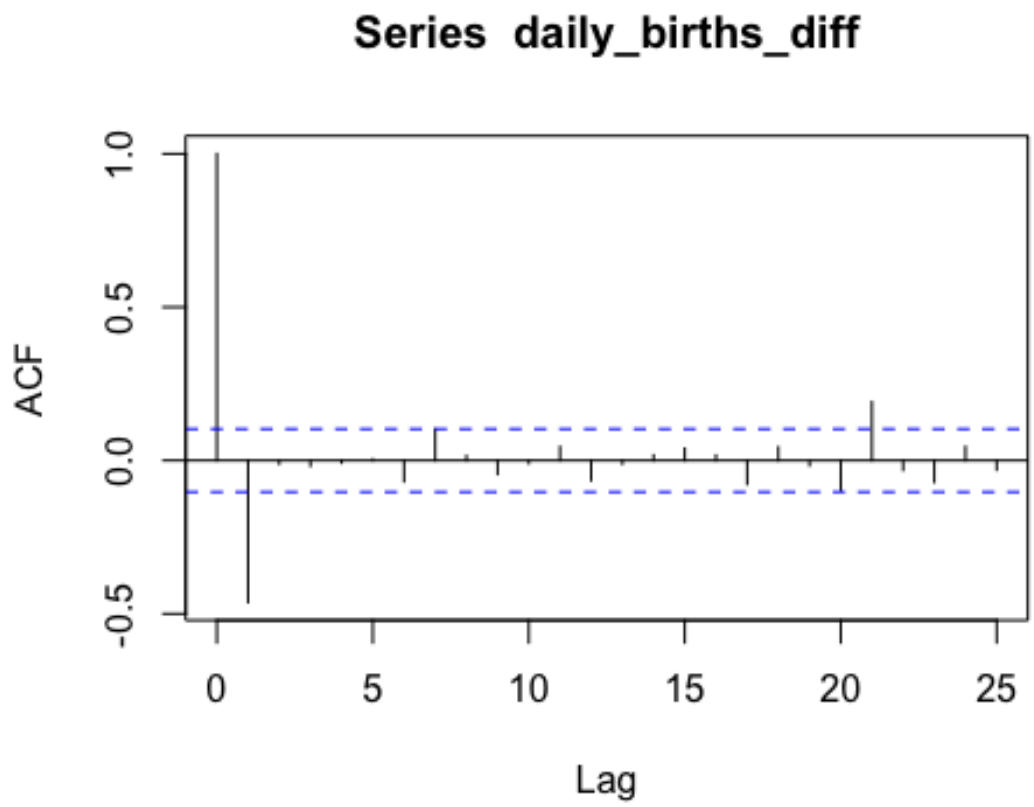
```
## Dickey-Fuller = -10.623, Lag order = 7, p-value = 0.01
## alternative hypothesis: stationary
```

```r
kpss.test(daily_births_diff)
```

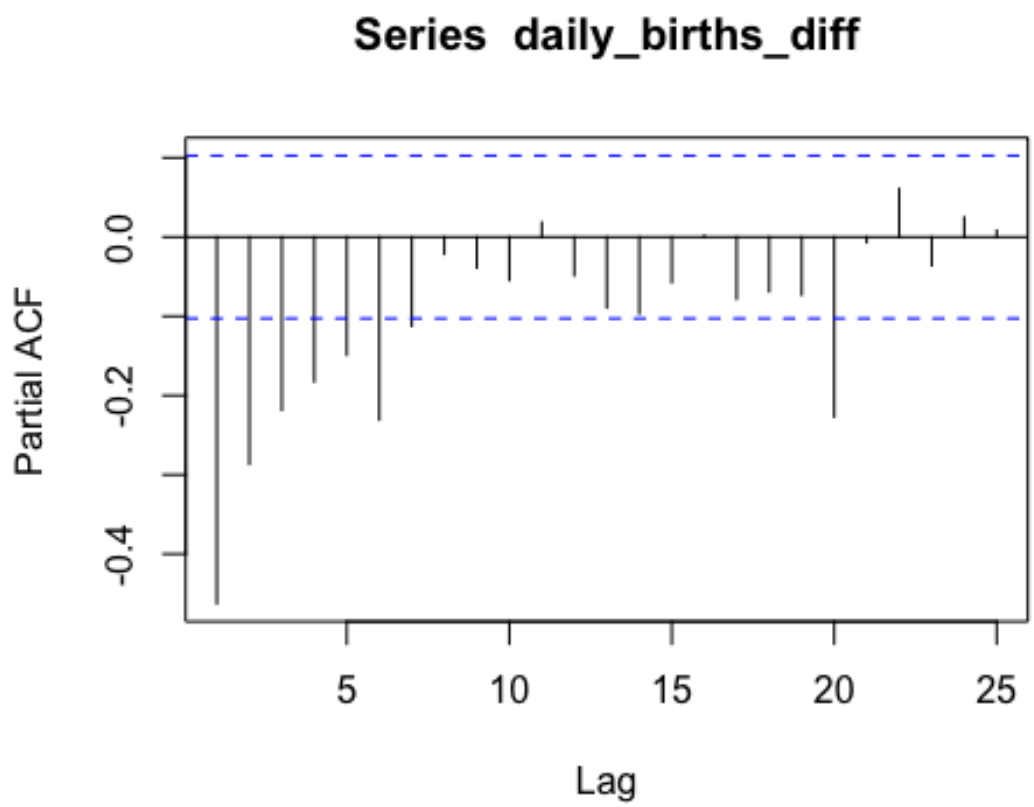
```
## Warning in kpss.test(daily_births_diff): p-value greater than printed p-
value

##
##  KPSS Test for Level Stationarity
##
## data:  daily_births_diff
## KPSS Level = 0.009464, Truncation lag parameter = 5, p-value = 0.1
```

ACF and PACF (after the first - order diff)

```r
acf(daily_births_diff)
```

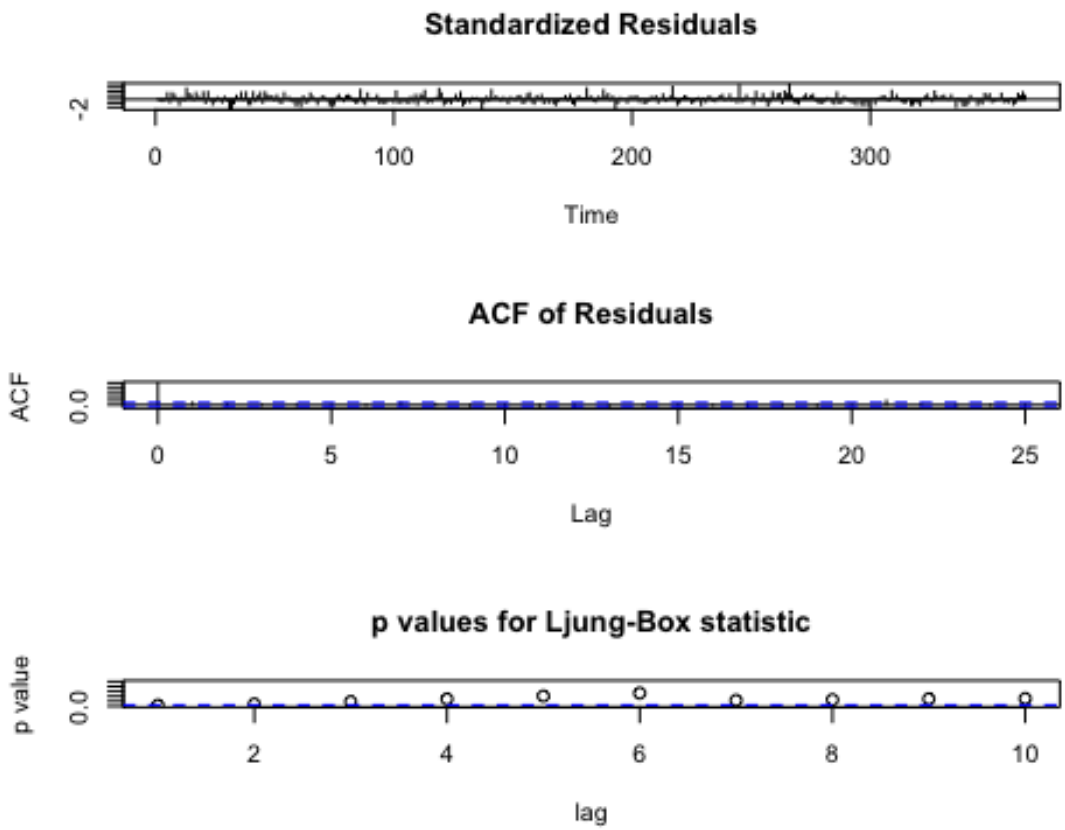



```r
pacf(daily_births_diff)
```

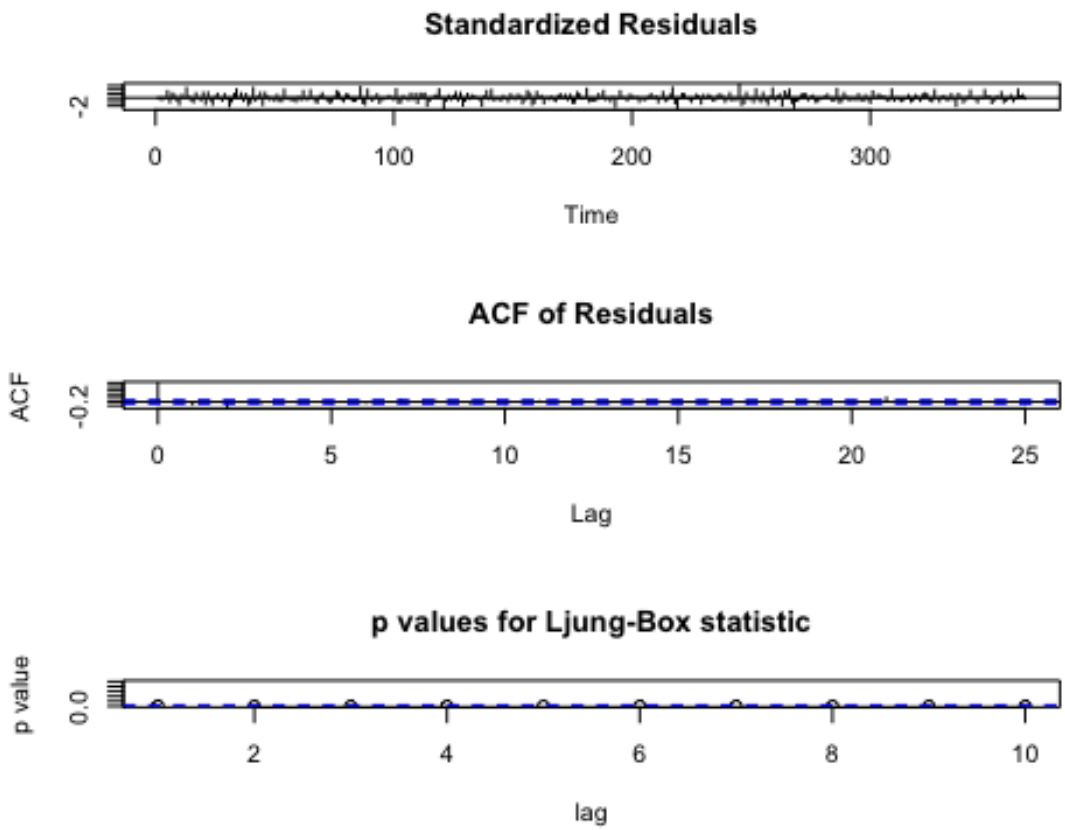


Choosing the model:

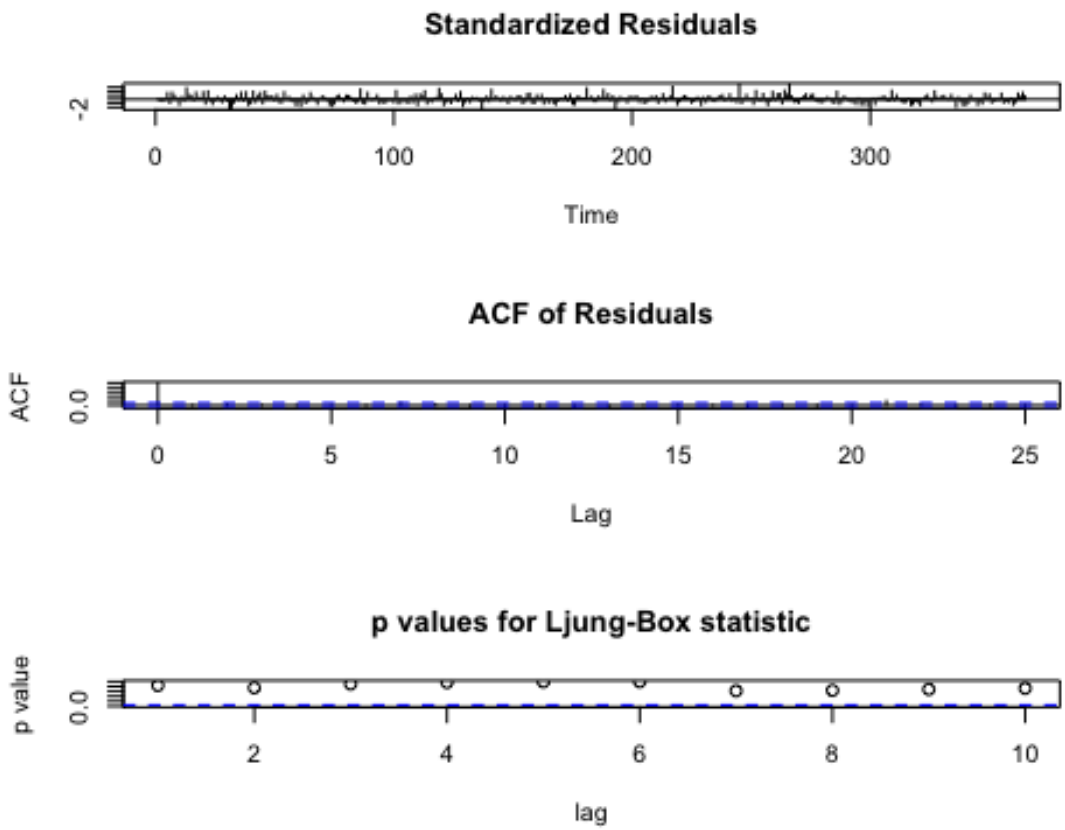
```
# ARIMA(0,1,1) , (1,1,0) , (1,1,1)

fit_011 <- arima(daily_births_ts, order = c(0,1,1))
fit_110 <- arima(daily_births_ts, order = c(1,1,0))
fit_111 <- arima(daily_births_ts, order = c(1,1,1))

AIC(fit_011, fit_110, fit_111)

##         df      AIC
## fit_011  2 2462.221
## fit_110  2 2563.261
## fit_111  3 2459.074

tsdiag(fit_011)
```




```
tsdiag(fit_110)
```

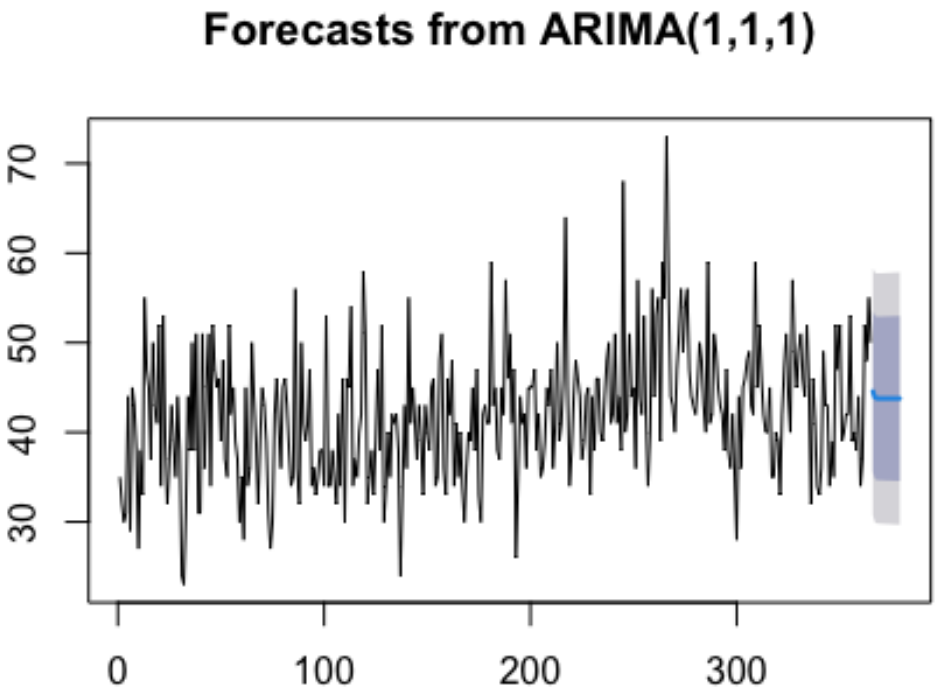



```
tsdiag(fit_111)
```



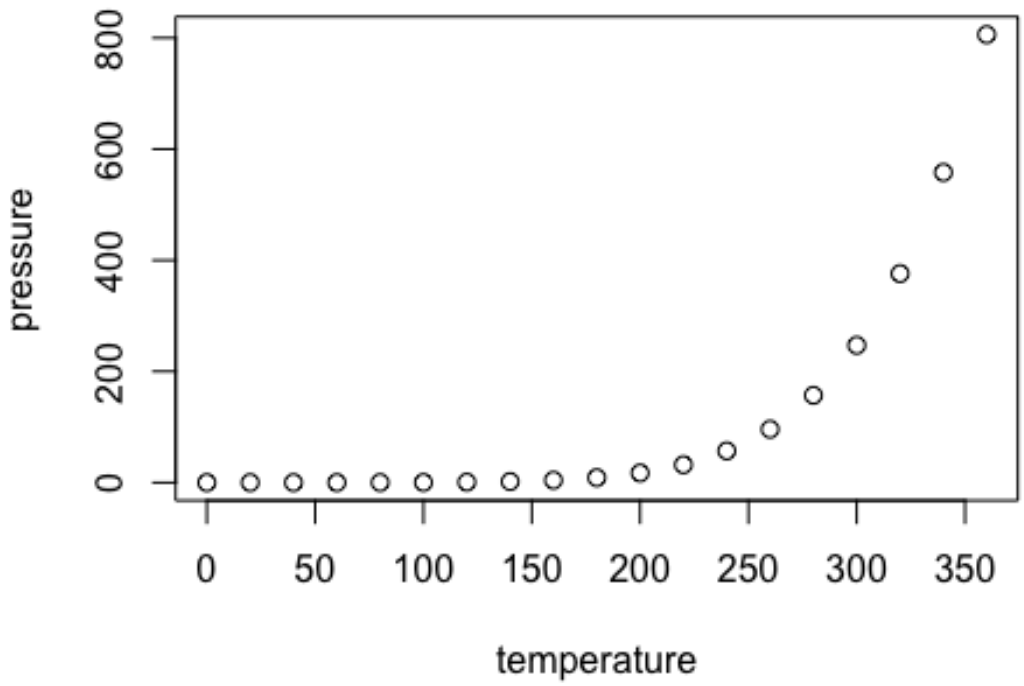
Forecast for ARIMA (1,1,1)

```
library(forecast)
fc <- forecast(fit_111, h = 14)
fc

##     Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## 366       44.55357 35.56918 53.53796 30.81314 58.29401
## 367       43.87142 34.76877 52.97407 29.95012 57.79272
## 368       43.78599 34.66847 52.90350 29.84195 57.73002
## 369       43.77528 34.64889 52.90168 29.81767 57.73290
## 370       43.77394 34.63932 52.90857 29.80375 57.74414
## 371       43.77378 34.63101 52.91654 29.79112 57.75643
## 372       43.77376 34.62286 52.92465 29.77867 57.76884
## 373       43.77375 34.61474 52.93276 29.76626 57.78125
## 374       43.77375 34.60663 52.94087 29.75385 57.79365
## 375       43.77375 34.59853 52.94898 29.74146 57.80605
## 376       43.77375 34.59043 52.95707 29.72907 57.81843
## 377       43.77375 34.58234 52.96516 29.71670 57.83080
## 378       43.77375 34.57426 52.97325 29.70434 57.84316
## 379       43.77375 34.56618 52.98132 29.69199 57.85551

plot(fc)
```



Including Plots

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

Time Series Project

Andreia Santos

Math 443 / Fall 2025

Dataset: *Daily total female births in California (1959)*

Analyze and forecast daily female birth counts in California using univariate time series methods.

The data come from a CSV file from research source kaggle.

It is based in a daily frequency, with one observation per day.

Shows day-to-day variability with no clear long-term trend; stationarity is formally assessed in later steps.

Project Overview

01

Exploratory Data Analysis
Stationarity Testing

02

Model Selection
Forecasting

03

Conclusion and
Recommendations

Exploratory Data Analysis

The raw time series plot shows daily fluctuations in female birth counts over the year.

Summary statistics (mean, standard deviation, minimum, and maximum) describe the overall level and variability of the series.

No clear long-term trend is observed, and seasonality cannot be reliably identified due to the one-year time span.

ACF and PACF plots are used to examine temporal dependence and guide ARIMA model selection.

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Trend and Seasonality

Trend

- No clear deterministic trend is observed.

Seasonality

- Seasonality cannot be reliably assessed due to the one-year time span of the data.

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Stationarity Testing (Original Series)

ADF and KPSS tests

- ADF suggests stationarity.
- KPSS rejects level stationarity.

Conclusion

- Tests provide mixed evidence.
- The series is treated as non-stationary as a conservative modeling choice.

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First-Order Differencing

Method

- First-order differencing:

  $$\Delta Y_t = Y_t - Y_{t-1}$$

Effect

- Stabilizes the mean.
- Produces a series fluctuating around zero.

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Stationarity Testing (Differenced Series)

ADF and KPSS tests after differencing

- Both tests indicate stationarity.

Conclusion

- One differencing step is sufficient.
- This validates d=1d = 1d=1.

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ACF and PACF (Model Identification)

ACF

- Strong spike at lag 1 with rapid decay.

PACF

- No sharp cutoff.

Interpretation

- Pattern is consistent with an MA(1) structure.

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Candidate ARIMA Models

Models considered

- ARIMA(0,1,1)
- ARIMA(1,1,0)
- ARIMA(1,1,1)

Goal

- Compare alternative dependence structures.

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Model Selection

Criterion

- Akaike Information Criterion (AIC).

Result

- ARIMA(1,1,1) has the lowest AIC.

Conclusion

- Best balance between model fit and complexity.

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Residual Diagnostics

Residual analysis

- No significant autocorrelation in residuals.
- Residuals behave like white noise.

Conclusion

- The selected model provides an adequate fit.

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Forecasting

Forecasting approach

- Short-term forecasts generated using ARIMA(1,1,1).

Uncertainty

- 80% and 95% prediction intervals reported.

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Forecast Interpretation

Key message

- The model does not predict sustained growth or decline.
- It predicts short-term stability.
- Forecast uncertainty increases as the horizon extends.

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Final Summary

Key findings

- The original series shows mixed stationarity evidence.
- First-order differencing achieves stationarity.
- ARIMA(1,1,1) is the preferred model.

Limitations

- Only one year of data.
- No explicit seasonal modeling.
- Long-horizon forecasts are highly uncertain.